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CONTROL DEVICE, OPERATION METHOD OF CONTROL DEVICE, OPERATION PROGRAM OF CONTROL DEVICE, IMAGING APPARATUS, AND DISPLAY APPARATUS

Abstract

A subject detection region setting unit and a display/storage region image generation unit of a CPU of a controller of a surveillance camera set a subject detection region where subject detection processing is performed and a display/storage region where display/storage processing is performed, for a captured image. The subject detection region setting unit switches the subject detection region between the display/storage region, which is a first region, and an entire surface region, which is a second region having a different size and position from the display/storage region.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS
[0001] The present application claims priority under 35 U.S.C. § 119 to Japanese Patent Application No. 2024-034377, filed on Mar. 6, 2024. The above application is hereby expressly incorporated by reference, in its entirety, into the present application.
BACKGROUND
1. Technical Field
[0002] The technology of the present disclosure relates to a control device, an operation method of a control device, an operation program of a control device, an imaging apparatus, and a display apparatus.
2. Description of the Related Art
[0003] An image processing apparatus described in JP2010-136204A includes an image input unit that inputs image data, a subject detection unit that detects a specific subject from the image data input by the image input unit, a region determination unit that determines a region where the detection by the subject detection unit is performed, and a recording unit that records the image data input by the image input unit on a recording medium. The region determination unit switches a determination method for a region where the detection of the subject is performed, according to whether or not moving image data is being recorded by the recording unit.
SUMMARY
[0004] One embodiment according to the technology of the present disclosure provides a control device, an operation method of a control device, an operation program of a control device, an imaging apparatus, and a display apparatus that are also capable of detecting a subject outside a region defined based on a specific region where specific processing is performed.
[0005] According to the present disclosure, there is provided a control device that performs subject detection processing and specific processing on a captured image obtained from an imaging element, the control device comprising: a processor, in which the processor is configured to: set a subject detection region where the subject detection processing is performed and a specific region where the specific processing is performed, for the captured image; and switch the subject detection region between a first region defined based on the specific region and a second region that is different from the first region in at least one of a size, a shape, or a position.
[0006] It is preferable that the second region has a larger region size than the first region.
[0007] It is preferable that the first region is the specific region, and the second region is an effective region of an imaging surface of the imaging element.
[0008] It is preferable that the processor is configured to: repeatedly switch the subject detection region between the first region and the second region.
[0009] It is preferable that the processor is configured to: switch the subject detection region to the second region in a case where a state in which no subject is detected continues for a first threshold value time after setting the subject detection region to the first region; and switch the subject detection region to the first region in a case where a state in which no subject is detected continues for a second threshold value time after setting the subject detection region to the second region.
[0010] It is preferable that the second threshold value time is different from the first threshold value time.
[0011] It is preferable that the second threshold value time is longer than the first threshold value time.

[0012] It is preferable that the control device is provided in a surveillance camera system including a pan mechanism, and the processor is configured to: switch the subject detection region from the second region to the first region in a case where the pan mechanism is operated while the subject detection region is the second region.
[0013] It is preferable that the processor is configured to: acquire brightness information indicating brightness of the first region and the second region; and switch the subject detection region to a region having higher brightness between the first region and the second region.
[0014] It is preferable that the processor is configured to: in a case where a subject is detected in the subject detection region, switch the subject detection region to a third region defined based on a detection result of the subject in a previous frame.
[0015] It is preferable that the third region is a region obtained by enlarging a region of the subject detected in the previous frame based on a set magnification factor, with a centroid of the region of the subject detected in the previous frame as a reference.
[0016] It is preferable that the processor is configured to: change the set magnification factor according to a movement speed of the detected subject in the captured image.
[0017] It is preferable that the processor is configured to: switch the subject detection region from the third region to the second region in a case where the subject detection region is set to the third region and a state in which no subject is detected continues for a third threshold value time.
[0018] It is preferable that the processor is configured to: perform, as the specific processing, at least one of processing related to display of an image of the specific region on a display unit or processing related to storage of the image of the specific region in a storage unit.
[0019] According to the present disclosure, there is provided an operation method of a control device that performs subject detection processing and specific processing on a captured image obtained from an imaging element, the operation method comprising: setting a subject detection region where the subject detection processing is performed and a specific region where the specific processing is performed, for the captured image; and switching the subject detection region between a first region defined based on the specific region and a second region that is different from the first region in at least one of a size, a shape, or a position.
[0020] According to the present disclosure, there is provided an operation program of a control device that performs subject detection processing and specific processing on a captured image obtained from an imaging element, the operation program causing a computer to perform a process comprising: setting a subject detection region where the subject detection processing is performed and a specific region where the specific processing is performed, for the captured image; and switching the subject detection region between a first region defined based on the specific region and a second region that is different from the first region in at least one of a size, a shape, or a position.
[0021] According to the present disclosure, there is provided an imaging apparatus comprising: the control device described above.
[0022] It is preferable that the imaging apparatus is a surveillance camera.
[0023] According to the present disclosure, there is provided a display apparatus comprising: the control device described above.
[0024] It is preferable that the display apparatus is a management apparatus of a surveillance camera system.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0025] Exemplary embodiments according to the technique of the present disclosure will be described in detail based on the following figures, wherein:

[0026] FIG. **1** is a diagram showing a surveillance camera system;
[0027] FIG. **2** is a block diagram showing a computer that constitutes a management apparatus;
[0028] FIG. **3** is a diagram showing an internal configuration of a surveillance camera;
[0029] FIG. **4** is a diagram showing an imaging element;
[0030] FIG. **5** is a block diagram showing a computer that constitutes a controller;
[0031] FIG. **6** is a block diagram of a CPU;
[0032] FIG. **7** is a diagram showing a threshold value time group;
[0033] FIG. **8** is a diagram showing subject detection region setting information;
[0034] FIG. **9** is a diagram showing a situation in which a subject detection region image is input to a subject detection model and subject detection information is output from the subject detection model;
[0035] FIG. **10** is a diagram showing a situation in which a distribution image is generated;
[0036] FIG. **11** is a diagram showing a transition in setting a subject detection region in a subject undetected state;
[0037] FIG. **12** is a diagram showing a transition in setting the subject detection region in a case where a subject is detected;
[0038] FIG. **13** is a diagram showing a detection frame enlargement region;
[0039] FIG. **14** is a diagram showing a transition in setting the subject detection region in a case where a subject non-detection state continues for a third threshold value time;
[0040] FIG. **15** is a diagram showing a case where a plurality of persons are detected;
[0041] FIG. **16** is a diagram showing a case where a person and a detection frame are partially outside a display/storage region;
[0042] FIG. **17** is a flowchart showing an operation procedure of the surveillance camera;
[0043] FIG. **18** is a flowchart showing an operation procedure of the surveillance camera;
[0044] FIG. **19** is a flowchart showing an operation procedure of the surveillance camera;
[0045] FIG. **20** is a flowchart showing an operation procedure of the surveillance camera;
[0046] FIG. **21** is a flowchart showing an operation procedure of the surveillance camera;
[0047] FIG. **22** is a diagram showing a transition in setting the subject detection region in a second embodiment;
[0048] FIG. **23** is a diagram showing a processing unit of a third embodiment;
[0049] FIG. **24** is a diagram showing a transition in setting the subject detection region in the third embodiment;
[0050] FIG. **25** is a diagram showing a processing unit of a fourth embodiment;
[0051] FIGS. **26**A and **26**B are diagrams showing a situation in which a set magnification factor is changed according to a movement speed of the subject, in which FIG. **26**A shows a case where the movement speed of the subject is less than a speed threshold value, and FIG. **26**B shows a case where the movement speed of the subject is equal to or greater than the speed threshold value; and
[0052] FIG. **27** is a diagram showing a management apparatus incorporating a control device.

DETAILED DESCRIPTION

First Embodiment

[0053] As shown in FIG. **1** as an example, a surveillance camera system **2** comprises a surveillance camera unit **10** and a management apparatus **11**. The surveillance camera unit **10** is installed at various surveillance locations such as streets, private homes, apartment buildings, public facilities, office buildings, factories, construction sites, and work sites. The surveillance camera unit **10** includes a surveillance camera **12**, a case **13**, a pan and tilt mechanism **14**, and a support post **15**. The surveillance camera **12** captures a video of a surveillance location to surveil whether there are any suspicious persons or any incidents or accidents. The surveillance camera **12** is an example of an “imaging apparatus” according to the technology of the present disclosure.
[0054] The surveillance camera **12** is accommodated in the case **13**. The case **13** is, for example, a highly sealed box-shaped container made of stainless steel and protects the surveillance camera **12**

from direct sunlight, wind and rain, and the like. A transmission window **16** is fitted into a front opening of the case **13**, and the surveillance camera **12** images the surveillance location through the transmission window **16**.

[0055] The pan and tilt mechanism **14** is attached to a center of a lower surface of the case **13**. The pan and tilt mechanism **14** revolves the case **13**, and consequently the surveillance camera **12**, by 360° endlessly in a horizontal direction in response to an operation by a user U of the surveillance camera system **2** through the management apparatus **11**. In addition, the pan and tilt mechanism **14** tilts the case **13**, and consequently the surveillance camera **12**, by +90° in a vertical direction in response to an operation by the user U through the management apparatus **11**. That is, an imaging region of the surveillance camera **12** can be changed in response to the operation by the user U. The pan and tilt mechanism **14** is an example of a “pan mechanism” according to the technology of the present disclosure.

[0056] The support post **15** is attached to a lower portion of the pan and tilt mechanism **14**. The support post **15** is attached to an appropriate position of the surveillance location such as a utility pole, a gate post, a ceiling, or a wall.

[0057] The surveillance camera unit **10** and the management apparatus **11** are connected to each other via a network **17** to communicate with each other. The network **17** is, for example, a wide area network (WAN) such as the Internet or a public communication network. The management apparatus **11** is installed in a control room or the like of a security company that comprehensively manages the surveillance camera system **2**. The management apparatus **11** is operated by the user U such as an employee of the security company. The management apparatus **11** is, for example, a desktop type personal computer and includes a display **18** and an input device **19** such as a keyboard, a mouse, a touch panel, and/or a microphone for audio input. The display **18** mainly displays an image obtained by imaging the surveillance location with the surveillance camera **12**, which is an image (distribution image **85** (refer to FIG. **6**)) distributed from the surveillance camera **12** via the network **17**. The display **18** is an example of a “display unit” according to the technology of the present disclosure. The input device **19** is operated by the user U to input various operation instructions. The operation instruction includes an imaging start instruction or an imaging end instruction of the surveillance camera **12**, an operation instruction for the pan and tilt mechanism **14**, a position change instruction (hereinafter, referred to as a zoom instruction) of a zoom lens **39** (refer to FIG. **3**), which will be described below, and the like.

[0058] As shown in FIG. **2** as an example, a computer that constitutes the management apparatus **11** comprises a storage **25**, a memory **26**, a central processing unit (CPU) **27**, and a communication unit **28**, in addition to the display **18** and the input device **19** mentioned above. The display **18**, the input device **19**, the storage **25**, the memory **26**, the CPU **27**, and the communication unit **28** are connected to each other via a busline **29**.

[0059] The storage **25** is a hard disk drive that is incorporated into the computer that constitutes the management apparatus **11** or that is connected to the computer through a cable or a network. Alternatively, the storage **25** is a disk array equipped with a plurality of hard disk drives connected in series. The storage **25** stores a control program such as an operating system, various application programs, various types of data associated with these programs, and the like. Additionally, the storage **25** stores the distribution image **85**. The storage **25** is an example of a “storage unit” according to the technology of the present disclosure. Note that a solid-state drive may be used instead of the hard disk drive.

[0060] The memory **26** is a work memory which is used for the CPU **27** to execute processing. The CPU **27** loads the program stored in the storage **25** into the memory **26** and executes processing in accordance with the program. Consequently, the CPU **27** comprehensively controls each unit of the computer. Note that the memory **26** may be incorporated into the CPU **27**. The communication unit **28** is a network interface that controls transmission of various types of information to and from the surveillance camera unit **10** via the network **17**.

[0061] As shown in FIG. **3** as an example, an imaging optical system **35** and an imaging element **36** are incorporated into the surveillance camera **12**. The imaging optical system **35** includes a plurality of types of lenses for forming an image on the imaging element **36**. Specifically, the imaging optical system **35** includes an objective lens **37**, a focus lens **38**, and the zoom lens **39**. These lenses **37** to **39** are disposed in this order from an object side toward an image formation side. Although simplified in FIG. **1**, each of the lenses **37** to **39** is actually a lens group composed of a plurality of lenses combined together.

[0062] The imaging optical system **35** also includes a stop **40**. The stop **40** is disposed on the image formation side with respect to the zoom lens **39**. The stop **40** is a so-called iris stop configured with a combination of a plurality of stop leaf blades.

[0063] A focus lens driving mechanism **41** is connected to the focus lens **38**, a zoom lens driving mechanism **42** is connected to the zoom lens **39**, and a stop driving mechanism **43** is connected to the stop **40**. The focus lens driving mechanism **41** includes a focus cam ring that holds the focus lens **38** and that includes a cam groove formed on an outer circumference thereof, a focus motor that rotates the focus cam ring around an optical axis OA to move the focus cam ring along the optical axis OA, a driver of the focus motor, and the like. Similarly, the zoom lens driving mechanism **42** includes a zoom cam ring that holds the zoom lens **39** and that includes a cam groove formed on an outer circumference thereof, a zoom motor that rotates the zoom cam ring around the optical axis OA to move the zoom cam ring along the optical axis OA, a driver of the zoom motor, and the like.

[0064] The stop **40** adjusts an amount of light passing through by simultaneously moving the plurality of stop leaf blades using a cam mechanism to open and close a central opening formed by inner edges of the plurality of stop leaf blades, that is, to change an opening degree of the opening. The stop driving mechanism **43** includes a stop motor that opens and closes the stop leaf blades, a driver of the stop motor, and the like.

[0065] The focus motor, the zoom motor, and the stop motor are, for example, stepping motors. In this case, a position of the focus lens **38** and a position of the zoom lens **39** on the optical axis OA and the opening degree of the stop **40** can be derived from driving amounts of the focus motor, the zoom motor, and the stop motor. Alternatively, instead of the driving amounts of the focus motor and the zoom motor, a position sensor may be provided to detect the position of the focus lens **38** and the position of the zoom lens **39**.

[0066] Electrical components such as the motor (the focus motor, the zoom motor, and the stop motor) or the driver of each of the driving mechanisms **41** to **43** are connected to a controller **44**. The electrical components of each of the driving mechanisms **41** to **43** are driven under the control of the controller **44**. For example, the controller **44** issues drive signals corresponding to various operation instructions input via the input device **19** of the management apparatus **11** to drive the electrical components of each of the driving mechanisms **41** to **43**. As an example, in a case where a zoom instruction to change an angle of view to a telephoto side is input, the controller **44** issues a drive signal to the driver of the zoom motor of the zoom lens driving mechanism **42** to move the zoom lens **39** to the telephoto side.

[0067] The focus motor, the zoom motor, and the stop motor output the driving amounts to the controller **44**. The controller **44** derives the position of the focus lens **38** and the position of the zoom lens **39** on the optical axis OA and the opening degree of the stop **40** from the driving amounts.

[0068] The imaging element **36** is, for example, a complementary metal-oxide-semiconductor (CMOS) image sensor. As shown in FIG. **4** as an example, the imaging element **36** includes a rectangular imaging surface **50** for capturing an image. The imaging surface **50** is formed by a plurality of pixels **51** arranged two-dimensionally. The pixel **51** accumulates signal charges corresponding to incident light and outputs an image signal (voltage signal) corresponding to the signal charges. The imaging element **36** is disposed such that a center of the imaging surface **50** matches the optical axis OA and the imaging surface **50** is orthogonal to the optical axis OA. It should be noted that the terms "match" and "orthogonal" used herein refer not only to completely matching and completely being orthogonal but also to matching and being orthogonal in a sense that includes an error generally acceptable within the technical field to which the technology of the present disclosure belongs.

[0069] Reference numeral **52** indicates a display/storage region. The display/storage region **52** is a region obtained by aligning a center thereof with a center of an entire surface region **53** (hereinafter, referred to as an entire surface region) of the imaging surface **50** and reducing the entire surface region **53** by a magnification set in advance. Therefore, it is obvious that the display/storage region **52** has a smaller region size than the entire surface region **53**. In addition, the entire surface region **53** has a different size and position from the display/storage region **52**. The display/storage region **52** is a region of the distribution image **85** that is distributed from the surveillance camera **12** to the management apparatus **11**, displayed on the display **18**, and stored in the storage **25**. The display/storage region **52** is an example of a "specific region" according to the technology of the present disclosure. Additionally, the entire surface region **53** is an example of an "effective region" according to the technology of the present disclosure. Although the entire surface region **53** is exemplified as the effective region, the technology of the present disclosure is not limited thereto. A region that is slightly smaller (for example, by several pixels) than the entire surface region **53**, specifically, a region that is preferably 80% or more, and more preferably 90% or more, of the entire surface region **53** may be the effective region.

[0070] In FIG. **3**, an imaging element driver **45** is connected to the imaging element **36**. The imaging element driver **45** is connected to the controller **44**. The imaging element driver **45** controls an image capturing timing of the imaging element **36** by supplying a vertical scanning signal and a horizontal scanning signal to the imaging element **36** and the like, under the control of the controller **44**. In addition, the imaging element driver **45** sets a gain corresponding to an International Organization for Standardization (ISO) sensitivity, which is a gain applied to the image signals output from the pixels **51**.

[0071] A connector **46** is further connected to the controller **44**. A cable for connection to the network **17** is attached to the connector **46**.

[0072] As shown in FIG. **5** as an example, the controller **44** is implemented by a computer that includes a storage **55**, a memory **56**, and a CPU **57**. The storage **55**, the memory **56**, and the CPU **57** are connected to each other via a busline **58**. The controller **44** is an example of a "computer" and a "control device" according to the technology of the present disclosure. The storage **55**, which is a non-transitory storage medium, is, for example, a non-volatile storage device such as an electrically erasable programmable read-only memory (EEPROM). The storage **55** stores various parameters and various programs. Alternatively, instead of the EEPROM, a ferroelectric random access memory (FeRAM) or a magnetoresistive random access memory (MRAM) may be used as the storage **55**.

[0073] The memory **56** is, for example, a random access memory (RAM) or the like and temporarily stores various types of information. The CPU **57** loads the program stored in the storage **55** into the memory **56** and executes processing in accordance with the program, thereby comprehensively controlling the operation of each unit of the surveillance camera **12**. The CPU **57** is an example of a "processor" according to the technology of the present disclosure. Note that the memory **56** may be incorporated into the CPU **57**.

[0074] As shown in FIG. **6** as an example, an operation program **65** is stored in the storage **55**. The operation program **65** is an application program for causing the computer that constitutes the controller **44** to function as the control device. That is, the operation program **65** is an example of an "operation program of a control device" according to the technology of the present disclosure. In addition to the operation program **65**, the storage **55** also stores a threshold value time group THG, display/storage region information **66**, a subject detection model **67**, and the like.

[0075] In a case where the operation program **65** is activated, the CPU **57** functions as an image processing unit **70**, a subject detection region setting unit **71**, a subject detection unit **72**, a timing unit **73**, a display/storage region image generation unit **74**, a distribution image generation unit **75**, and a distribution control unit **76** in cooperation with the memory **56** and the like.
[0076] The image processing unit **70** acquires a captured image **80** output from the imaging element **36**. The captured image **80** is composed of image signals of all the pixels **51** of the imaging element **36**, that is, image signals of the entire surface region **53**. The image processing unit **70** performs various types of image processing on the captured image **80**. The various types of image processing are, for example, offset correction processing, sensitivity correction processing, pixel interpolation processing, white balance correction processing, gamma correction processing, demosaicing processing, brightness signal and color difference signal generation processing, contour enhancement processing, color correction processing, and the like. The image processing unit **70** outputs the captured image **80** after various types of image processing to the subject detection region setting unit **71** and the display/storage region image generation unit **74**.
[0077] The subject detection region setting unit **71** sets a region (hereinafter, referred to as a subject detection region) where a subject is detected by the subject detection unit **72**. The threshold value time group THG is input to the subject detection region setting unit **71**. As shown in FIG. **7** as an example, the threshold value time group THG includes a first threshold value time TH**1**, a second threshold value time TH**2**, and a third threshold value time TH**3**.
[0078] The subject detection region setting unit **71** extracts the portion of the subject detection region from the captured image **80** to generate a subject detection region image **90** (refer to FIG. **8**). The subject detection region setting unit **71** outputs subject detection region setting information **81** including the subject detection region image **90** to the subject detection unit **72** and the timing unit **73**. As shown in FIG. **8** as an example, the subject detection region setting information **81** includes the set subject detection region in addition to the subject detection region image **90**.
[0079] The subject detection unit **72** performs subject detection processing. That is, the subject detection unit **72** uses the subject detection model **67** to detect the subject present in the subject detection region set by the subject detection region setting unit **71**. More specifically, as shown in FIG. **9** as an example, the subject detection unit **72** inputs the subject detection region image **90** to the subject detection model **67** and causes the subject detection model **67** to output subject detection information **82**. The subject detection model **67** is, for example, a machine learning model such as a convolutional neural network. The subject detection information **82** is an example of a “detection result of the subject” according to the technology of the present disclosure.
[0080] As shown in the drawing, in a case where a target subject, here, a person P, is captured in the subject detection region image **90**, position coordinates of a rectangular detection frame (referred to as a bounding box) **92** surrounding the person P are registered in the subject detection information **82**. On the other hand, in a case where no person P is captured in the subject detection region image **90**, nothing is registered in the subject detection information **82**. The detection frame **92** is an example of a “region of the subject detected” according to the technology of the present disclosure. The subject detection unit **72** outputs the subject detection information **82** to the subject detection region setting unit **71** and the distribution image generation unit **75**. In FIG. **9**, for convenience of description, the person P is depicted with a dashed line in the subject detection information **82**. The same applies to subsequent figures such as FIG. **10**.
[0081] The timing unit **73** measures various times to be compared with the threshold value time group THG. The timing unit **73** outputs timing information **83** to the subject detection region setting unit **71**.
[0082] The display/storage region information **66** is information for defining the display/storage region **52** and is, for example, position coordinates of two pixels **51** diagonally located at an upper left corner and a lower right corner of the display/storage region **52**. The display/storage region image generation unit **74** performs display/storage processing based on the display/storage region

information **66**. That is, the display/storage region image generation unit **74** generates a display/storage region image **84** by extracting the portion of the display/storage region **52** from the captured image **80** based on the display/storage region information **66**. In this way, the processing of generating the display/storage region image **84** by the display/storage region image generation unit **74** is an example of "specific processing", "processing related to display", and "processing related to storage" according to the technology of the present disclosure. Additionally, the display/storage region image **84** is an example of an "image of a specific region" according to the technology of the present disclosure. The display/storage region image generation unit **74** outputs the display/storage region image **84** to the distribution image generation unit **75**.

[0083] The distribution image generation unit **75** generates the distribution image **85** based on the subject detection information **82** and the display/storage region image **84**. More specifically, as shown in FIG. **10**, the distribution image generation unit **75** generates the distribution image **85** by combining the detection frame **92** registered in the subject detection information **82** with the display/storage region image **84**. The distribution image generation unit **75** outputs the distribution image **85** to the distribution control unit **76**. The distribution control unit **76** performs control of distributing the distribution image **85** to the management apparatus **11**.

[0084] Each of the processing units **70** to **76** performs each processing described above, each time the captured image **80** is input from the imaging element **36**. Meanwhile, a frame rate of the captured image **80** is, for example, 15 to 60 frames per second (fps).

[0085] As shown in FIG. **11** as an example, in a case where the imaging start instruction of the surveillance camera **12** is issued by the user U via the input device **19** and the imaging of the surveillance location by the surveillance camera **12** is started, the subject detection region setting unit **71** sets the subject detection region to the entire surface region **53**. In this case, the entire surface region **53** is registered in the subject detection region of the subject detection region setting information **81** as illustrated in FIG. **8**. In addition, the captured image **80** is registered in the subject detection region image **90** of the subject detection region setting information **81**. The entire surface region **53** is an example of a "second region" according to the technology of the present disclosure.

[0086] The timing unit **73** measures a time for which the subject detection region is set to the entire surface region **53** by the subject detection region setting unit **71** (hereinafter, referred to as a setting time of the entire surface region **53**) and outputs the timing information **83** to the subject detection region setting unit **71** at predetermined intervals. The subject detection region setting unit **71** compares the setting time of the entire surface region **53** of the timing information **83** with the second threshold value time TH**2**. Then, in a case where the setting time of the entire surface region **53** continues for the second threshold value time TH**2**, the subject detection region is switched from the entire surface region **53** to the display/storage region **52**. The second threshold value time TH**2** is, for example, 1 minute. In this case, the display/storage region **52** is registered in the subject detection region of the subject detection region setting information **81**. In addition, the display/storage region image **84** is registered in the subject detection region image **90** of the subject detection region setting information **81**. The display/storage region **52** is an example of a "first region" according to the technology of the present disclosure.

[0087] The timing unit **73** measures a time for which the subject detection region is set to the display/storage region **52** by the subject detection region setting unit **71** (hereinafter, referred to as a setting time of the display/storage region **52**) and outputs the timing information **83** to the subject detection region setting unit **71** at predetermined intervals. The subject detection region setting unit **71** compares the setting time of the display/storage region **52** of the timing information **83** with the first threshold value time TH**1**. Then, in a case where the setting time of the display/storage region **52** continues for the first threshold value time TH**1**, the subject detection region is switched again from the display/storage region **52** to the entire surface region **53**. The first threshold value time TH**1** is shorter than the second threshold value time TH**2** and is, for example, 45 seconds. The subject detection region setting unit **71** repeatedly switches the subject detection region alternately between the display/storage region **52** and the entire surface region **53** as long as a state in which no subject is detected by the subject detection unit **72** (hereinafter, referred to as a subject undetected state) continues.

[0088] As shown in FIG. **12** as an example, in a case where the subject detection region is set to the display/storage region **52** or the entire surface region **53** and the subject is detected by the subject detection unit **72**, the subject detection region setting unit **71** switches the subject detection region from the display/storage region **52** or the entire surface region **53** to a detection frame enlargement region **95** (refer to FIG. **13**). A maximum value of a size of the detection frame enlargement region **95** is the entire surface region **53**. In FIG. **12**, a case where the display/storage region **52** is switched to the detection frame enlargement region **95** is illustrated. The detection frame enlargement region **95** is an example of a “third region” according to the technology of the present disclosure. In the following description, a case where the subject is detected by the subject detection unit **72** will be referred to as a subject detection state.

[0089] As shown in FIG. **13** as an example, the detection frame enlargement region **95** is a region obtained by enlarging the detection frame **92** based on the set magnification factor, with a center CF of the detection frame **92** of the subject detected in a previous frame as a reference. Here, the detection frame enlargement region **95** is a region where vertical sides of the detection frame **92** are scaled by 1.25 times and horizontal sides are scaled by 2.5 times. In this case, 1.25 times and 2.5 times are examples of a “set magnification factor” according to the technology of the present disclosure. Additionally, the center CF of the detection frame **92** is an example of a “centroid of the region of the subject” according to the technology of the present disclosure. As can be seen from the example of FIG. **13**, the detection frame enlargement region **95** may protrude from the display/storage region **52** depending on the position of the original detection frame **92**.

[0090] In a case where the person P is moving, the position and/or the size of the detection frame **92** changes with the movement of the person P. Therefore, the position and/or the size of the detection frame enlargement region **95** set based on the detection frame **92** also changes with the movement of the person P.

[0091] As shown in FIG. **14** as an example, in a case where the subject detection region is set to the detection frame enlargement region **95** and a state in which no subject is detected by the subject detection unit **72** (hereinafter, referred to as a subject non-detection state) occurs, the timing unit **73** measures a duration of the subject non-detection state and outputs the timing information **83** to the subject detection region setting unit **71** at predetermined intervals. The subject detection region setting unit **71** compares the duration of the subject non-detection state of the timing information **83** with the third threshold value time TH**3**. Then, in a case where the subject non-detection state continues for the third threshold value time TH**3**, the subject detection region is switched from the detection frame enlargement region **95** to the entire surface region **53**. The subject non-detection state occurs in a case where the entire body of the person P is no longer captured in the captured image **80** (in a case where the entire body of the person P moves outside the entire surface region **53**). During the period from when the subject non-detection state occurs until the third threshold value time TH**3** has elapsed, the subject detection region setting unit **71** continues to set the subject detection region to the detection frame enlargement region **95** set immediately before the subject non-detection state occurs. The third threshold value time TH**3** is shorter than the first threshold value time TH**1** and the second threshold value time TH**2** and is, for example, 5 seconds. After switching the subject detection region from the detection frame enlargement region **95** to the entire surface region **53**, the subject detection region setting unit **71** repeatedly switches the subject detection region between the display/storage region **52** and the entire surface region **53**.

[0092] Here, the subject undetected state can be rephrased as a state from when the imaging of the surveillance location by the surveillance camera **12** is started until the subject is first detected by the subject detection unit **72**. In addition, the subject undetected state can be rephrased as a state from when the subject detection region is switched from the detection frame enlargement region **95** to the entire surface region **53**, after the subject detection region has been set to the detection frame enlargement region **95** and the subject non-detection state has continued for the third threshold value time TH**3** or longer, until the subject is detected again by the subject detection unit **72**.

[0093] In FIG. **9**, a case where only one person P is captured in the subject detection region image **90** has been illustrated, but the technology of the present disclosure is not limited thereto. As shown in FIG. **15** as an example, in a case where a plurality of persons P, here, three persons P**1**, P**2**, and P**3**, are captured in the subject detection region image **90**, the subject detection model **67** detects the persons P**1** to P**3** and outputs the subject detection information **82** in which the position coordinates of the detection frames **92** of the persons P**1** to P**3** are registered. In such a case, the subject detection region setting unit **71** sets the detection frame enlargement region **95** for each of the persons P**1** to P**3**. Therefore, the detection frame enlargement region **95** may partially overlap with another detection frame enlargement region **95**.

[0094] In addition, in FIG. **10**, a case where the person P and the detection frame **92** fit within the display/storage region **52** has been illustrated, but the technology of the present disclosure is not limited thereto. As shown in FIG. **16** as an example, the person P and the detection frame **92** may be partially outside the display/storage region **52**. For example, in a case where only the head of the person P is captured in the display/storage region **52**, the detection frame **92** may be changed to a small frame that surrounds only the head of the person P rather than the entire body of the person P.

[0095] Next, operations of the above-described configuration will be described with reference to flowcharts shown in FIGS. **17**, **18**, **19**, **20**, and **21** as an example. In a case where the operation program **65** is activated in the surveillance camera **12**, as shown in FIG. **6**, the CPU **57** of the controller **44** of the surveillance camera **12** functions as the image processing unit **70**, the subject detection region setting unit **71**, the subject detection unit **72**, the timing unit **73**, the display/storage region image generation unit **74**, the distribution image generation unit **75**, and the distribution control unit **76**.

[0096] In the management apparatus **11**, in a case where the imaging start instruction of the surveillance camera **12** is issued by the user U via the input device **19** and the imaging of the surveillance location by the surveillance camera **12** is started, the captured image **80** of the surveillance location is output from the imaging element **36**. The captured image **80** is input to the image processing unit **70**, and various types of image processing are performed by the image processing unit **70** (step ST**100** in FIG. **17**). The captured image **80** after various types of image processing is output from the image processing unit **70** to the subject detection region setting unit **71** and the display/storage region image generation unit **74**.

[0097] In a case where the state of the surveillance camera **12** is the subject undetected state (YES in step ST**110**), subject detection region setting processing in the subject undetected state is performed by the subject detection region setting unit **71** (step ST**120**). Details of the subject detection region setting processing in the subject undetected state will be described with reference to FIGS. **18** and **19**. On the other hand, in a case where the state of the surveillance camera **12** is not the subject undetected state (NO in step ST**110**), the processing transitions to step ST**130**.

[0098] In a case where the subject is detected in the previous frame by the subject detection unit **72**, that is, in a case where the previous frame is in the subject detection state (YES in step ST**130**), as shown in FIGS. **12** and **13**, the subject detection region setting unit **71** sets the subject detection region to the detection frame enlargement region **95** obtained by enlarging the detection frame **92** based on the set magnification factor, with the center CF of the detection frame **92** of the subject detected in the previous frame as a reference (step ST**140**). On the other hand, in a case where no subject is detected in the previous frame by the subject detection unit **72**, that is, in a case where the previous frame is in the subject non-detection state (NO in step ST**130**), the subject detection region setting unit **71** sets the subject detection region to the detection frame enlargement region **95** set immediately before the subject non-detection state occurs (step ST**150**).

[0099] In a case where the subject detection region is set by the subject detection region setting unit **71**, the subject detection region setting information **81** shown in FIG. **8** is generated. The subject detection region setting information **81** is output from the subject detection region setting unit **71** to the subject detection unit **72** and the timing unit **73**.
[0100] The subject present in the subject detection region is detected by the subject detection unit **72** (step ST**160**). More specifically, as shown in FIG. **9**, the subject detection region image **90** is input to the subject detection model **67**, and the subject detection information **82** is output from the subject detection model **67**. The subject detection information **82** is output from the subject detection unit **72** to the subject detection region setting unit **71** and the distribution image generation unit **75**.
[0101] The display/storage region image generation unit **74** generates the display/storage region image **84** by extracting the portion of the display/storage region **52** from the captured image **80** based on the display/storage region information **66**. The display/storage region image **84** is output from the display/storage region image generation unit **74** to the distribution image generation unit **75**.
[0102] As shown in FIG. **10**, the distribution image generation unit **75** generates the distribution image **85** by combining the detection frame **92** registered in the subject detection information **82** with the display/storage region image **84**. The distribution image **85** is output from the distribution image generation unit **75** to the distribution control unit **76** and is distributed to the management apparatus **11** under the control of the distribution control unit **76** (step ST**170**).
[0103] Subsequently, subject detection state determination processing is performed by the subject detection region setting unit **71** (step ST**180**). Details of the subject detection state determination processing will be described with reference to FIGS. **20** and **21**.
[0104] Each processing of steps ST**100** to ST**180** is repeatedly performed until the imaging end instruction of the surveillance camera **12** is issued by the user U via the input device **19** and the imaging of the surveillance location by the surveillance camera **12** has not ended (NO in step ST**190**), in the management apparatus **11**.
[0105] FIGS. **18** and **19** show a procedure of the subject detection region setting processing in the subject undetected state in step ST**120**. First, in a case where it is the first frame immediately after the imaging of the surveillance location by the surveillance camera **12** is started (YES in step ST**12001**), the subject detection region is set to the entire surface region **53** by the subject detection region setting unit **71** as shown in FIG. **11** (step ST**12002**). Then, the timing unit **73** starts the time measurement of the setting time of the entire surface region **53** (step ST**12003**).
[0106] In a case where it is the second or subsequent frame (NO in step ST**12001**), the subject detection region setting unit **71** compares the setting time of the entire surface region **53** of the timing information **83** with the second threshold value time TH**2** and determines whether or not the setting time of the entire surface region **53** continues for the second threshold value time TH**2** (step ST**12004**). In a case where the setting time of the entire surface region **53** does not continue for the second threshold value time TH**2** (NO in step ST**12004**), the subject detection region is set to the entire surface region **53** by the subject detection region setting unit **71** (step ST**12005**). On the other hand, in a case where the setting time of the entire surface region **53** continues for the second threshold value time TH**2** (YES in step ST**12004**), the processing transitions to step ST**12006**.
[0107] In step ST**12006**, the subject detection region setting unit **71** compares the setting time of the display/storage region **52** of the timing information **83** with the first threshold value time TH**1** and determines whether or not the setting time of the display/storage region **52** continues for the first threshold value time TH**1**. In a case where the setting time of the display/storage region **52** continues for the first threshold value time TH**1** (YES in step ST**12006**), the subject detection region is set to the entire surface region **53** by the subject detection region setting unit **71** (step ST**12007**). In other words, the subject detection region is switched from the display/storage region **52** to the entire surface region **53**. Then, the timing unit **73** resets the setting time of the entire

surface region **53** of the timing information **83** to zero and starts the time measurement of the setting time of the entire surface region **53** (step ST**12008**). Additionally, the time measurement of the setting time of the display/storage region **52** is stopped, and the setting time of the display/storage region **52** of the timing information **83** is reset to zero (step ST**12009**).

[0108] In a case where the setting time of the display/storage region **52** does not continue for the first threshold value time TH**1** (NO in step ST**12006**), the subject detection region is set to the display/storage region **52** by the subject detection region setting unit **71** (step ST**12010**). In other words, the subject detection region is switched from the entire surface region **53** to the display/storage region **52**. Then, the timing unit **73** starts the time measurement of the setting time of the display/storage region **52** (step ST**12011**).

[0109] FIGS. **20** and **21** show a procedure of the subject detection state determination processing of step ST**180**. First, in a case where the subject is detected in step ST**160** (YES in step ST**18001**), the timing unit **73** resets the duration of the subject non-detection state of the timing information **83** to zero (step ST**18002**).

[0110] In a case where the subject detected in step ST**160** is the subject detected from the subject undetected state (YES in step ST**18003**), the subject detection region setting unit **71** determines that the state of the surveillance camera **12** has transitioned from the subject undetected state to the subject detection state (step ST**18004**). On the other hand, in a case where the subject detected in step ST**160** is not the subject detected from the subject undetected state (NO in step ST**18003**), the subject detection region setting unit **71** determines that the state of the surveillance camera **12** is the subject detection state (step ST**18005**). A case where the subject detected in step ST**160** is not the subject detected from the subject undetected state refers to a case where the subject detection region is set to the detection frame enlargement region **95** and the subject is captured in the detection frame enlargement region **95**.

[0111] In a case where no subject is detected in step ST**160** (NO in step ST**18001**), the processing transitions to step ST**18006**. In step ST**18006**, the subject detection region setting unit **71** determines whether or not the state of the surveillance camera **12** is the subject undetected state. In a case where the state of the surveillance camera **12** is not the subject undetected state (NO in step ST**18006**), the timing unit **73** measures the duration of the subject non-detection state (step ST**18007**). A case where no subject is detected in step ST**160** and the state of the surveillance camera **12** is not the subject undetected state refers to a case where the subject detection region is set to the detection frame enlargement region **95** and no subject is captured in the detection frame enlargement region **95**.

[0112] In subsequent step ST**18008**, the subject detection region setting unit **71** compares the duration of the subject non-detection state of the timing information **83** with the third threshold value time TH**3** and determines whether or not the subject non-detection state continues for the third threshold value time TH**3**. In a case where the subject non-detection state continues for the third threshold value time TH**3** (YES in step ST**18008**), the subject detection region setting unit **71** determines that the state of the surveillance camera **12** has transitioned from the subject detection state to the subject undetected state (step ST**18009**). On the other hand, in a case where the subject non-detection state does not continue for the third threshold value time TH**3** (NO in step ST**18008**), the subject detection region setting unit **71** determines that the state of the surveillance camera **12** is the subject detection state (step ST**18010**). The state (subject detection state or subject undetected state) of the surveillance camera **12** determined in the subject detection state determination processing of step ST**180** is used for the determination of step ST**110** of the next frame.

[0113] As described above, the subject detection region setting unit **71** and the display/storage region image generation unit **74** of the CPU **57** of the controller **44** of the surveillance camera **12** set the subject detection region where the subject detection processing is performed and the display/storage region **52** where the display/storage processing is performed, for the captured image **80**. The subject detection region setting unit **71** switches the subject detection region between the display/storage region **52**, which is the first region, and the entire surface region **53**, which is the second region having a different size and position from the display/storage region **52**. Therefore, it is also possible to detect the subject outside the display/storage region **52**.

[0114] As shown in FIG. **4**, the entire surface region **53** has a larger region size than the display/storage region **52**. Therefore, it is possible to detect the subject present in the entire surface region **53** having a larger region size than the display/storage region **52**. In addition, even for the same subject, the subject is captured relatively larger in the display/storage region **52** than in the entire surface region **53**. Therefore, in a case where the subject detection region is set to the display/storage region **52**, the subject can be detected with higher accuracy than in a case where the subject detection region is set to the entire surface region **53**.

[0115] The first region is the display/storage region **52**, and the second region is the entire surface region **53**. Therefore, it is possible to meet both the need to detect the subject present in the display/storage region **52** with high accuracy and the need to preemptively detect the subject present in the entire surface region **53** outside the display/storage region **52**.

[0116] As shown in FIG. **11**, the subject detection region setting unit **71** repeatedly switches the subject detection region between the display/storage region **52** and the entire surface region **53**. Therefore, it is possible to increase the opportunities to meet both the need to detect the subject present in the display/storage region **52** with high accuracy and the need to preemptively detect the subject present in the entire surface region **53** outside the display/storage region **52**.

[0117] As shown in FIG. **11**, the subject detection region setting unit **71** switches the subject detection region to the entire surface region **53** in a case where a state in which no subject is detected continues for the first threshold value time TH**1** after setting the subject detection region to the display/storage region **52**. In addition, in a case where a state in which no subject is detected continues for the second threshold value time TH**2** after the subject detection region is set to the entire surface region **53**, the subject detection region is switched to the display/storage region **52**. Therefore, it is possible to increase the opportunities to meet both the need to detect the subject present in the display/storage region **52** with high accuracy and the need to preemptively detect the subject present in the entire surface region **53** outside the display/storage region **52**.

[0118] As shown in FIG. **11**, the second threshold value time TH**2** is different from the first threshold value time TH**1**. Specifically, the second threshold value time TH**2** is longer than the first threshold value time TH**1**. Therefore, it is possible to increase the opportunities to meet the need to preemptively detect the subject present in the entire surface region **53** outside the display/storage region **52**. In a case where the priority is given to the need to detect the subject present in the display/storage region **52** with high accuracy rather than to the need to preemptively detect the subject present in the entire surface region **53** outside the display/storage region **52**, the first threshold value time TH**1** may be set to be longer than the second threshold value time TH**2**.

[0119] As shown in FIGS. **12** and **13**, the subject detection region setting unit **71** switches the subject detection region to the detection frame enlargement region **95** defined based on the subject detection information **82** in the previous frame in a case where the subject is detected by the subject detection unit **72**. In the detection frame enlargement region **95** defined based on the subject detection information **82** in the previous frame, the probability that the subject will still be present in the detection frame enlargement region **95** in the next frame is high. Additionally, even for the same subject, the subject is captured relatively larger in the detection frame enlargement region **95** than in the entire surface region **53**. Therefore, the subject can be detected with high accuracy.

[0120] As shown in FIG. **13**, the detection frame enlargement region **95** is a region obtained by enlarging the detection frame **92** based on the set magnification factor, with the center CF of the detection frame **92** surrounding the subject detected in the previous frame as a reference. Therefore, the detection frame enlargement region **95** can be easily set.

[0121] As shown in FIG. **14**, the subject detection region setting unit **71** switches the subject detection region from the detection frame enlargement region **95** to the entire surface region **53** in a case where the subject detection region is set to the detection frame enlargement region **95** and a state in which no subject is detected continues for the third threshold value time TH**3**. Therefore, in a case where the subject is no longer detected, it is possible to automatically return to a state in which the subject present in the entire surface region **53** outside the display/storage region **52** is preemptively detected.

[0122] The display/storage region image generation unit **74** performs processing of generating the display/storage region image **84** as the specific processing. Therefore, the distribution image **85** based on the display/storage region image **84** can be displayed on the display **18** or stored in the storage **25**.

[0123] The surveillance camera **12** has a high necessity to detect a subject such as a suspicious person in order to perform a role of surveilling the surveillance location. Therefore, it is possible to greatly exhibit the effect of being able to detect the subject outside the display/storage region **52**.

Second Embodiment

[0124] As shown in FIG. **22** as an example, in a second embodiment, the subject detection region setting unit **71** switches the subject detection region from the entire surface region **53** to the display/storage region **52** in a case where the user U operates the pan and tilt mechanism **14** while the subject detection region is the entire surface region **53**. The subject detection region setting unit **71** resets the subject detection region to the entire surface region **53** in a case where the setting time of the display/storage region **52** continues for a fourth threshold value time TH**4**. The fourth threshold value time TH**4** is stored in the storage **55** and is, for example, 1 minute.

[0125] A case where the user U operates the pan and tilt mechanism **14** mainly refers to a case where a subject of interest to the user U is present in a direction in which the pan and tilt mechanism **14** is operated. Therefore, in a case where the pan and tilt mechanism **14** is operated by the user U, the subject detection region is switched to the display/storage region **52** having a higher detection accuracy of the subject than the entire surface region **53**. Accordingly, the subject of interest to the user U, which is present in the direction in which the pan and tilt mechanism **14** is operated, can be detected with high accuracy.

[0126] In a case where the zoom instruction of the surveillance camera **12** is issued by the user U, the subject detection region may be switched from the entire surface region **53** to the display/storage region **52**.

Third Embodiment

[0127] As shown in FIG. **23** as an example, in a third embodiment, the CPU **57** functions as a brightness detection unit **100** in addition to the processing units **70** to **76**. The brightness detection unit **100** detects brightness of each of the display/storage region **52** and the entire surface region **53** at predetermined intervals. The brightness is, for example, an average value of brightness values obtained from the pixel values of the pixels **51** that constitute the display/storage region **52** and the entire surface region **53**. The brightness detection unit **100** outputs brightness information **101** indicating the detected brightness of the display/storage region **52** and the entire surface region **53** to the subject detection region setting unit **71**.

[0128] As shown in FIG. **24** as an example, the subject detection region setting unit **71** switches the subject detection region to a region having higher brightness between the display/storage region **52** and the entire surface region **53**. FIG. **24** illustrates a case where the subject detection region is switched to the entire surface region **53** because the entire surface region **53** is brighter than the display/storage region **52**, as a result of the brightness being detected by the brightness detection unit **100** while the subject detection region is set to the display/storage region **52**. In this case, the subject can always be detected in the region having higher brightness, where subject detection accuracy is higher. For example, in a case where the display/storage region **52** is not illuminated with streetlights, but the entire surface region **53** outside the display/storage region **52** is illuminated with streetlights, switching the subject detection region to the brighter entire surface region **53** allows the detection of the subject present in the entire surface region **53** illuminated with the streetlights.

[0129] As can be seen from the second embodiment and the third embodiment described above, an aspect of repeatedly switching the subject detection region between the display/storage region **52** and the entire surface region **53** as in the first embodiment is not essential in the technology of the present disclosure. The subject detection region may be set to the entire surface region **53** by default, and the subject detection region may be switched to the display/storage region **52** only in a case where the pan and tilt mechanism **14** is operated by the user U. In addition, the subject detection region may be set to the entire surface region **53** by default, and the subject detection region may be switched to the display/storage region **52** only in a case where the display/storage region **52** is brighter than the entire surface region **53**. The second embodiment and the third embodiment described above may be implemented in combination. Obviously, the first embodiment, the second embodiment, and/or the third embodiment described above may be implemented in combination.

Fourth Embodiment

[0130] As shown in FIG. **25** as an example, in a fourth embodiment, the CPU **57** functions as a subject speed detection unit **105** in addition to the processing units **70** to **76**. The subject speed detection unit **105** detects a movement speed of the subject detected by the subject detection unit **72** in the captured image **80**. The subject speed detection unit **105** detects the movement speed of the subject based on, for example, the amount of change in the position of the center CF of the detection frame **92** in two consecutive frames. The subject speed detection unit **105** outputs subject speed information **106** indicating the detected movement speed of the subject to the subject detection region setting unit **71**.

[0131] A speed threshold value STH is input to the subject detection region setting unit **71**. The speed threshold value STH is stored in the storage **55**. The subject detection region setting unit **71** compares the movement speed of the subject indicated by the subject speed information **106** with the speed threshold value STH. Then, as shown in FIG. **26**A as an example, in a case where the movement speed of the subject is less than the speed threshold value STH, the subject detection region setting unit **71** sets the detection frame enlargement region **95** by enlarging the detection frame **92** based on the set magnification factor (vertical 1.25 times, horizontal 2.5 times) shown in FIG. **13** of the above-described first embodiment. Meanwhile, in a case where the movement speed of the subject is equal to or greater than the speed threshold value STH, the subject detection region setting unit **71** sets the detection frame enlargement region **95** by enlarging the detection frame **92** based on the set magnification factor (vertical 2 times, horizontal 4 times) that is larger than the set magnification factor (vertical 1.25 times, horizontal 2.5 times) shown in FIG. **13** of the above-described first embodiment. That is, the subject detection region setting unit **71** changes the set magnification factor of the detection frame enlargement region **95** according to the detected movement speed of the subject in the captured image **80**.

[0132] In a case where the movement speed is equal to or greater than the speed threshold value STH, there is a concern that the subject in the current frame may move outside the detection frame enlargement region **95** set based on the detection frame **92** in the previous frame. Therefore, in a case where the movement speed is equal to or greater than the speed threshold value STH, the subject detection region setting unit **71** sets the detection frame enlargement region **95** at the set magnification factor that is larger than the normal set magnification factor, making it possible to reduce the likelihood of the subject moving outside the detection frame enlargement region **95** and the subject being missed in the detection.

[0133] The movement speed of the subject in the captured image **80** includes not only a movement speed in a case where the subject itself walks or runs but also a movement speed in a case where the subject appears to move due to a change in the position of the subject in the captured image **80** when the pan and tilt mechanism **14** is operated.

[0134] In a case where a state in which the movement speed is equal to or greater than the speed threshold value STH continues for a set number of frames, the set magnification factor may be changed to a larger value. In addition, in a case where a state in which the movement speed is less than the speed threshold value STH continues for the set number of frames after the set magnification factor is changed to a large value, the set magnification factor may be returned to the original small value.

[0135] The set magnification factor may be changed in three or more stages, such as setting the set magnification factor to a value smaller than the normal value in a case where the movement speed of the subject is less than a first speed threshold value, setting the set magnification factor to the normal value in a case where the movement speed of the subject is equal to or greater than the first speed threshold value and less than a second speed threshold value, and setting the set magnification factor to a value larger than the normal value in a case where the movement speed of the subject is equal to or greater than the second speed threshold value.

[0136] The "processing related to display" may be processing of performing control of displaying the display/storage region image **84** on a display unit such as the display **18**. Similarly, the "processing related to storage" may be processing of performing control of storing the display/storage region image **84** in a storage unit such as the storage **25**.

[0137] The "specific processing" includes not only the "processing related to display" and the "processing related to storage" described as an example, but also all processing performed on the captured image **80** other than the subject detection processing. In addition, the "specific processing" may not be the same processing for each frame.

[0138] In a case where the display/storage region **52** is set as the subject detection region and the subject detection region image **90** generated by the subject detection region setting unit **71** is the display/storage region image **84**, the display/storage region image **84** may be output from the subject detection region setting unit **71** to the distribution image generation unit **75** and used to generate the distribution image **85**. In this case, the display/storage region image generation unit **74** is not operated.

[0139] The centroid of the region of the subject is not limited to the center CF of the detection frame **92** described as an example. The centroid of the region of the subject may be a centroid of a polygonal shape surrounding the subject.

[0140] Although the display/storage region **52** is exemplified as the first region, the technology of the present disclosure is not limited thereto. Any of a display region on the display **18** or a storage region in the storage **25** may be used as the first region. The display region also includes a cutout region by electronic zoom. Additionally, a region that is slightly smaller or larger (for example, by several pixels) than the display/storage region **52** may be used as the first region. Further, the detection frame enlargement region **95** has been exemplified as the third region, but the technology of the present disclosure is not limited thereto. The detection frame **92** itself may be used as the third region.

[0141] Although a case where the display/storage region **52**, which is the first region, and the entire surface region **53**, which is the second region, are both rectangular and have the same shape has been exemplified, the technology of the present disclosure is not limited thereto. The first region and the second region may have different shapes, such as the first region being circular or elliptical and the second region being rectangular.

[0142] The display/storage region **52** may not be a region having the same size, shape, and position in each frame, and for example, the size, shape, and position may be different for each set frame.

[0143] The subject is not limited to the person P described as an example. The subject may be an animal such as a deer, a wild boar, a monkey, a bear, or a bird, or a vehicle such as a car, a train, or an airplane. It is preferable to change the set magnification factor of the detection frame enlargement region **95** according to the type of the subject, such as making the horizontal magnification larger than the vertical magnification in a case where the subject is a car.

[0144] The subject has been detected using the subject detection model **67**, but the technology of the present disclosure is not limited thereto. The subject may be detected using well-known pattern recognition techniques.

[0145] In each of the above-described embodiments, the surveillance camera **12** has been described as an example of an "imaging apparatus" according to the technology of the present disclosure, but the technology of the present disclosure is not limited thereto. Instead of the surveillance camera **12**, a digital camera, a smartphone, a tablet terminal, or the like used by a general user may be used. In addition, an electronic endoscope may be used.

[0146] In each of the above-described embodiments, an aspect in which the controller **44** corresponding to the "control device" according to the technology of the present disclosure is provided in the surveillance camera **12** has been exemplified, but the technology of the present disclosure is not limited thereto. As shown in FIG. **27** as an example, the "control device" according to the technology of the present disclosure may be provided in the management apparatus **11**. Alternatively, the "control device" according to the technology of the present disclosure may have functions provided through distribution between the management apparatus **11** and the surveillance camera **12**.

[0147] In each of the above-described embodiments, for example, as a hardware structure of processing units that execute various types of processing, such as the image processing unit **70**, the subject detection region setting unit **71**, the subject detection unit **72**, the timing unit **73**, the display/storage region image generation unit **74**, the distribution image generation unit **75**, the distribution control unit **76**, the brightness detection unit **100**, and the subject speed detection unit **105**, the following various processors can be used. The various processors include, as mentioned above, as well as a general-purpose processor such as the CPU **57**, which executes software (operation program **65**) to function as various processing units, a programmable logic device (PLD) such as a field programmable gate array (FPGA), which is a processor whose circuit configuration can be changed after manufacturing, a dedicated electrical circuit having a custom-designed circuit configuration for executing specific processing such as an application-specific integrated circuit (ASIC), and the like.

[0148] One processing unit may be composed of one of these various processors or composed of a combination of two or more processors of the same or different types (for example, a combination of a plurality of FPGAs and/or a combination of a CPU and an FPGA). Additionally, a plurality of processing units may be configured with a single processor.

[0149] As an example of configuring a plurality of processing units with a single processor, first, there is an aspect in which one or more CPUs are combined with software to configure a single processor, and the processor functions as a plurality of processing units, as represented by computers such as clients and servers. Second, there is an aspect in which a processor is used to implement the functions of the entire system, including a plurality of processing units, with a single integrated circuit (IC) chip, as represented by a system-on-chip (SoC) and the like. In this way, the various processing units are configured using one or more of the above-described various processors as the hardware structure.

[0150] Further, as the hardware structure of these various processors, more specifically, electrical circuitry obtained by combining circuit elements, such as semiconductor elements, can be used.

[0151] From the above description, the technology described in the following supplementary claims can be understood.

Supplementary Claim 1

[0152] A control device that performs subject detection processing and specific processing on a captured image obtained from an imaging element, the control device comprising: [0153] a processor, [0154] in which the processor is configured to: [0155] set a subject detection region where the subject detection processing is performed and a specific region where the specific processing is performed, for the captured image; and [0156] switch the subject detection region between a first region defined based on the specific region and a second region that is different from the first region in at least one of a size, a shape, or a position.

Supplementary Claim 2

[0157] The control device according to Supplementary claim 1, [0158] in which the second region has a larger region size than the first region.

Supplementary Claim 3

[0159] The control device according to Supplementary claim 2, [0160] in which the first region is the specific region, and the second region is an effective region of an imaging surface of the imaging element.

Supplementary Claim 4

[0161] The control device according to any one of Supplementary claims 1 to 3, [0162] in which the processor is configured to: [0163] repeatedly switch the subject detection region between the first region and the second region.

Supplementary Claim 5

[0164] The control device according to Supplementary claim 4, [0165] in which the processor is configured to: [0166] switch the subject detection region to the second region in a case where a state in which no subject is detected continues for a first threshold value time after setting the subject detection region to the first region; and [0167] switch the subject detection region to the first region in a case where a state in which no subject is detected continues for a second threshold value time after setting the subject detection region to the second region.

Supplementary Claim 6

[0168] The control device according to Supplementary claim 5, [0169] in which the second threshold value time is different from the first threshold value time.

Supplementary Claim 7

[0170] The control device according to Supplementary claim 6, [0171] in which the second threshold value time is longer than the first threshold value time.

Supplementary Claim 8

[0172] The control device according to any one of Supplementary claims 1 to 7, [0173] in which the control device is provided in a surveillance camera system including a pan mechanism, and [0174] the processor is configured to: [0175] switch the subject detection region from the second region to the first region in a case where the pan mechanism is operated while the subject detection region is the second region.

Supplementary Claim 9

[0176] The control device according to any one of Supplementary claims 1 to 8, [0177] in which the processor is configured to: [0178] acquire brightness information indicating brightness of the first region and the second region; and [0179] switch the subject detection region to a region having higher brightness between the first region and the second region.

Supplementary Claim 10

[0180] The control device according to any one of Supplementary claims 1 to 9, [0181] in which the processor is configured to: [0182] in a case where a subject is detected in the subject detection region, switch the subject detection region to a third region defined based on a detection result of the subject in a previous frame.

Supplementary Claim 11

[0183] The control device according to Supplementary claim 10, [0184] in which the third region is a region obtained by enlarging a region of the subject detected in the previous frame based on a set magnification factor, with a centroid of the region of the subject detected in the previous frame as a reference.

Supplementary Claim 12

[0185] The control device according to Supplementary claim 11, [0186] in which the processor is configured to: [0187] change the set magnification factor according to a movement speed of the detected subject in the captured image.

Supplementary Claim 13
[0188] The control device according to any one of Supplementary claims 10 to 12, [0189] in which the processor is configured to: [0190] switch the subject detection region from the third region to the second region in a case where the subject detection region is set to the third region and a state in which no subject is detected continues for a third threshold value time.
Supplementary Claim 14
[0191] The control device according to any one of Supplementary claims 1 to 13, [0192] in which the processor is configured to: [0193] perform, as the specific processing, at least one of processing related to display of an image of the specific region on a display unit or processing related to storage of the image of the specific region in a storage unit.
Supplementary Claim 15
[0194] An imaging apparatus comprising: [0195] the control device according to any one of Supplementary claims 1 to 14.
Supplementary Claim 16
[0196] The imaging apparatus according to Supplementary claim 15, which is a surveillance camera.
Supplementary Claim 17
[0197] A display apparatus comprising: [0198] the control device according to any one of Supplementary claims 1 to 14.
Supplementary Claim 18
[0199] The display apparatus according to Supplementary claim 17, which is a management apparatus of a surveillance camera system.
[0200] The technology of the present disclosure can also be appropriately combined with various embodiments and/or various modification examples mentioned above. Additionally, it is obvious that the technology of the present disclosure is not limited to each of the above-described embodiments and various configurations can be employed without departing from the gist. Furthermore, the technology of the present disclosure extends not only to the program but also to a storage medium that stores the program in a non-transitory manner.
[0201] The described contents and the shown contents provided above are detailed descriptions of portions related to the technology of the present disclosure and are merely examples of the technology of the present disclosure. For example, the descriptions regarding the above-described configuration, function, operation, and effect are descriptions regarding one example of the configuration, function, operation, and effect of the portions related to the technology of the present disclosure. Therefore, it goes without saying that, within the scope that does not depart from the gist of the technology of the present disclosure, unnecessary portions may be deleted, new elements may be added, or replacement may be made with respect to the described contents and the shown contents provided above. Additionally, in order to avoid confusion and facilitate understanding of the portions related to the technology of the present disclosure, descriptions regarding common general knowledge and the like that do not require special descriptions for enabling the implementation of the technology of the present disclosure have been omitted from the described contents and the shown contents provided above.
[0202] In this specification, “A and/or B” is synonymous with “at least one of A or B”. That is, “A and/or B” means that A may be used alone, B may be used alone, or a combination of A and B may be used. Moreover, in this specification, the same concept as “A and/or B” is also applied to a case where three or more matters are expressed by “and/or”.
[0203] All documents, patent applications, and technical standards described in this specification are incorporated in this specification by reference to the same extent as in a case where each individual document, patent application, and technical standard were specifically and individually stated to be incorporated by reference.

Claims

1. A control device comprising: a processor, wherein the processor is configured to: set a subject detection region where a subject detection processing is performed and a specific region where a specific processing is performed, for the captured image; and switch the subject detection region between a first region defined based on the specific region and a second region that is different from the first region in at least one of a size, a shape, or a position.
2. The control device according to claim 1, wherein the second region has a larger region size than the first region.
3. The control device according to claim 2, wherein the first region is the specific region, and the second region is an effective region of an imaging surface of an imaging element.
4. The control device according to claim 1, wherein the processor is configured to: repeatedly switch the subject detection region between the first region and the second region.
5. The control device according to claim 4, wherein the processor is configured to: switch the subject detection region to the second region in a case where a state in which no subject is detected continues for a first threshold value time after setting the subject detection region to the first region; and switch the subject detection region to the first region in a case where a state in which no subject is detected continues for a second threshold value time after setting the subject detection region to the second region.
6. The control device according to claim 5, wherein the second threshold value time is different from the first threshold value time.
7. The control device according to claim 6, wherein the second threshold value time is longer than the first threshold value time.
8. The control device according to claim 1, wherein the control device is provided in a surveillance camera system including a pan mechanism, and the processor is configured to: switch the subject detection region from the second region to the first region in a case where the pan mechanism is operated while the subject detection region is the second region.
9. The control device according to claim 1, wherein the processor is configured to: acquire brightness information indicating brightness of the first region and the second region; and switch the subject detection region to a region having higher brightness between the first region and the second region.
10. The control device according to claim 1, wherein the processor is configured to: in a case where a subject is detected in the subject detection region, switch the subject detection region to a third region defined based on a detection result of the subject in a previous frame.
11. The control device according to claim 10, wherein the third region is a region obtained by enlarging a region of the subject detected in the previous frame based on a set magnification factor, with a centroid of the region of the subject detected in the previous frame as a reference.
12. The control device according to claim 11, wherein the processor is configured to: change the set magnification factor according to a movement speed of the detected subject in the captured image.
13. The control device according to claim 10, wherein the processor is configured to: switch the subject detection region from the third region to the second region in a case where the subject detection region is set to the third region and a state in which no subject is detected continues for a third threshold value time.
14. The control device according to claim 1, wherein the processor is configured to: perform, as the specific processing, at least one of processing related to display of an image of the specific region on a display unit or processing related to storage of the image of the specific region in a storage unit.
15. An operation method of a control device, the operation method comprising: setting a subject detection region where a subject detection processing is performed and a specific region where a

specific processing is performed, for a captured image; and switching the subject detection region between a first region defined based on the specific region and a second region that is different from the first region in at least one of a size, a shape, or a position.

16. A non-transitory computer-readable storage medium storing an operation program causing a computer to perform a process comprising: setting a subject detection region where a subject detection processing is performed and a specific region where a specific processing is performed, for a captured image; and switching the subject detection region between a first region defined based on the specific region and a second region that is different from the first region in at least one of a size, a shape, or a position.

17. An imaging apparatus comprising: the control device according to claim 1.

18. The imaging apparatus according to claim 17, which is a surveillance camera.

19. A display apparatus comprising: the control device according to claim 1.

20. The display apparatus according to claim 19, which is a management apparatus of a surveillance camera system.